



Instructions:

- This question paper comprises of 2 questions.
- All questions are mandatory.
- Course outcomes () and marks [] are listed to extreme right against each of the questions

- Q.1 Fig. 1 shows the scenario in the Scipher-Eggli-Sandoz (SES) protocol for causally ordered unicast. Apply the SES protocol for the causal ordering of messages sent by nodes N_1 , N_2 and N_3 as shown in Fig. 1. A message m_1 is sent by N_1 to N_3 , m_2 is sent by N_1 to N_2 , and m_5 is sent by N_2 to N_3 . Messages m_3 and m_4 are sent by N_3 to N_1 . Elaborate all the steps of the protocol including the contents of the vectors maintained by the nodes. Can the protocol be used to keep the replicas of a node consistent if different updates are applied to these replicas at different times? (CO2) [9M]

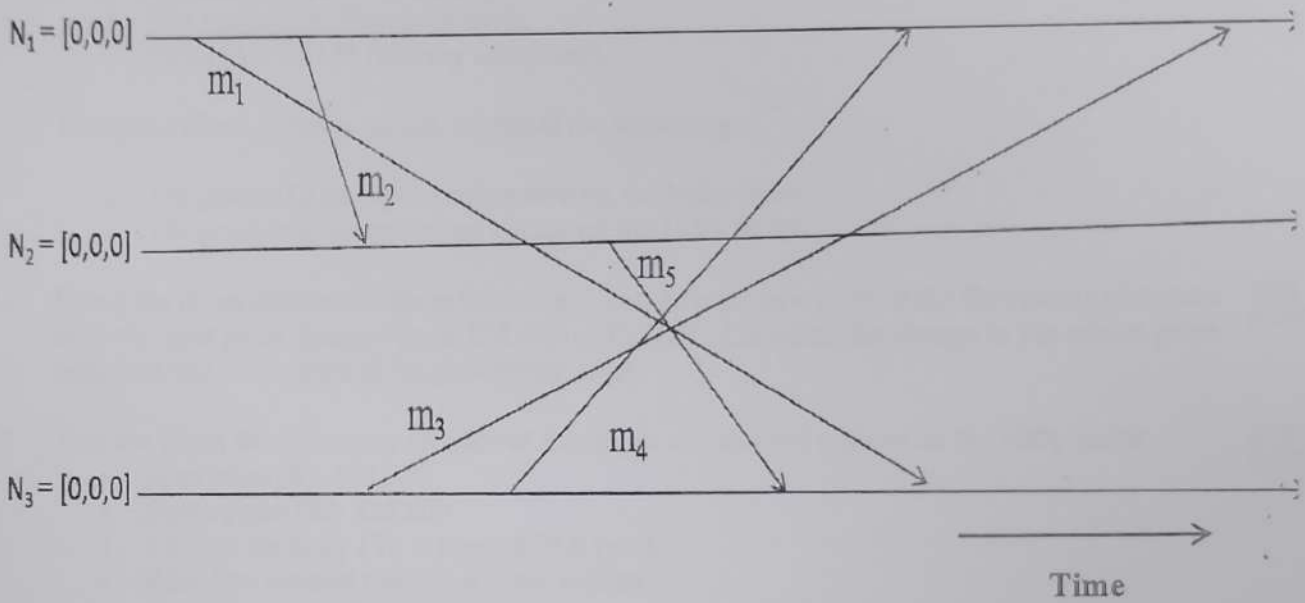


Fig. 1

- Q.2 Answer the following w.r.t. Remote Procedure Calls. (CO1) [3M]
- (a). How would you implement “at least once” and “at most once” semantics? [3M]
- (b). What are the challenges for the implementation of “at most once” semantics? [3M]